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PERFORMANCE SPECIFICATION

SPEED PICKUP, MAGNETIC (METRIC)

This specification is approved for use by the U.S. Army Tank Automotive Research, Development and Engineering Center (TARDEC), Research, Development and Engineering Command (RDECOM), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance and inspection requirements for magnetic speed pickups hereinafter referred to as pickups (see 6.1).

1.2 Classification. The pickups covered by this specification are of the following types, as specified (see 6.2).

Type I - High pressure compressor speed (N_H) in accordance with Drawing 12286861
Type II - Power turbine speed (N_P) in accordance with Drawing 12286840

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRNAS MS#268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to DAMI_STANDARDIZATION@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

COMMERCIAL ITEM DESCRIPTIONS

A-A-59133 - Cleaning Compound, High Pressure (Steam) Cleaner

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-810 - Environmental Test Methods

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

US ARMY TACOM DRAWINGS

12302067 - Speed Pickup, Magnetic
12302350 - Speed Pickup, Electromagnetic
12286861 - Speed Pickup, Electromagnetic
12286840 - Adapter, Speedometer-Tachometer

(Copies of these drawings are available from DAMI_STANDARDIZATION@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS#268, 6501 E. 11 Mile Road, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

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AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available from www.asq.org or American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials used for the pickup shall be in accordance with the manufacturer's materials specifications. The materials shall be capable of meeting all of the operational and environmental requirements specified herein (see 4.4.1).

3.2.1 Corrosion resistance. The pickup shall not be affected by exposure to military-type fuels, lubricating oils, cleaning solvents, and salt water (see 4.4.1).

3.2.2 Potting. If potting is used within the pickup, the cross-sectional area exposed to the external environment shall be kept to an absolute minimum (see 4.4.1).

3.2.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs (see 4.4.1).

3.3 Design and construction. The pickup shall meet the following design parameters (see 4.4.2):

a. Minimum wire diameter shall be 0.0711 millimeter (mm) [#41 American Wire Gauge (AWG)].
b. Minimum unloaded or unstressed wall thickness shall be 0.508 mm [0.020 inch (in)]. This does not pertain to the thickness of the front face shield on blind ended pickups which is limited only by standard design practice.
c. Minimum loaded, stressed or threaded wall thickness (taken from the maximum inside diameter to the root of the external thread) shall be 0.762 mm (0.030 in).

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3.3.1 Envelope. Envelope dimensions for each type of pickup shall be in accordance with the requirements of the applicable drawing (see 4.4.2):

Type	Drawing Number
I	12302067
II	12302350

3.3.2 Weight. The weight of the pickup shall be as follows (see 4.4.2):

Type	Size
I	0.15 kilograms (kg) (0.33 pounds (lb))
II	0.16 kg (0.35 lb)

3.4 Performance.

3.4.1 Insulation resistance. The minimum insulation resistance between the coil and case shall be 1 megohm at 50 volts (V) root mean square (rms) (see 4.4.3).

3.4.2 Dielectric strength. Dielectric strength shall be sufficient for the pickup to sustain 100 V rms applied between the coil and case (see 4.4.4).

3.4.3 Output voltage. The output voltage shall be in accordance with table I for the applicable type of pickup (see 4.4.5).

TABLE I. Output voltage.

Pickup type	Engine speed (%)	Output voltage (peak)	Load (ohms)	Surface speed (mmps [ips]) 1/	Frequency (pps) 2/	Test gap mm (in)	Coil resistance (ohms)
Type I	5 100	0.3 5.0	4700 4700	3352 (132) 66700 (2626)	66 1313	2.54 (0.1) 2.54 (0.1)	85-115 85-115
Type II	5 54 108	0.4 4.0 7.0	4700 4700 4700	2261 (89) 24562 (967) 49124 (1934)	45 483 967	2.54 (0.1) 2.54 (0.1) 2.54 (0.1)	80-460 80-460 80-460

1/ (mmps [ips]) = millimeters per second (inches per second).
2/ pps = pulses per second.

3.4.4 Phasing. The pickup shall be wired so that pin A is positive with respect to pin B when the pickup tip is approached by a magnetic object (see 4.4.6).

3.5 Environmental conditions.

3.5.1 Temperature. The pickup shall operate when exposed to temperatures of -54 to 204 degrees Celsius (°C) (-65 to 400 degrees Fahrenheit (°F)) as specified in 4.4.7.1.

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3.5.2 Altitude. The pickup shall operate when exposed to altitudes of 396 meters (m) (1,300 feet (ft)) below sea level to 3048 m (10,000 ft) above sea level (see 4.4.7.2).

3.5.3 Humidity. The pickup shall show no evidence of corrosion or other defects that would affect the function or structural strength when exposed to zero to 100 percent (%) relative humidity as specified in 4.4.7.3.

3.5.4 Salt fog. The pickup shall operate when exposed to a salt-fog as specified in 4.4.7.4.

3.5.5 Sand and dust. The pickup shall operate after 6 hours exposure to blowing dust as specified in 4.4.7.5.

3.5.6 Fungus. The pickup shall show no evidence of microbial growth that would adversely affect performance after 28 days exposure to a spore incubation as specified in 4.4.7.6.

3.5.7 Chemical resistance. The pickup shall meet the specified performance requirements when exposed to fuel vapors, diesel fuel oil, military combat gasoline, aviation turbine fluid, solvents, and steam cleaning fluid (A-A-59133) in operating, non-operating and transit conditions, as specified in 4.4.7.7.

3.5.8 Vibration. The pickup shall meet the specified performance requirements when exposed to vibration having levels up to 5 gravity units (g) from 5 to 2000 Hertz (Hz) as specified in 4.4.7.8.

3.5.9 Shock. The pickup shall not be damaged when subjected to the shock specified in 4.4.7.9.

3.5.9.1 Gun firing shock. The pickup shall meet the specified performance requirements without malfunction when exposed to shocks of 100±10 g for 1.5±0.2 millisecond (ms) duration in any direction as specified in 4.4.7.9.1.

3.5.9.2 Ballistic shock. The pickup shall meet the specified performance requirements without malfunction when exposed to half sine wave shock pulses of 200±20 g in any direction as specified in 4.4.7.9.2.

3.5.9.3 Basic shock. The pickup shall meet the specified performance requirements without malfunction when exposed to half sine wave shock pulses of 40 g for 18 ms duration in any direction (see 4.4.7.9.3).

3.5.10 Submergence. The pickup shall operate without short- or long-term damage after being submerged in the fluid (see 3.5.7) at a pressure of 345 kilopascals (kPa) (50 pounds per square inch gage (psig)) for a period of 2 hours (see 4.4.7.10).

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3.6 Identification marking. Identification marking shall be in accordance with MIL-STD-130 (see 4.4.2 and 6.2).

3.7 Workmanship. The pickup shall be fabricated and finished to meet all criteria for appearance, fit, and print tolerance specifications. Particular attention shall be given to wiring, marking, plating and keeping parts free of any burrs or sharp edges (see 4.4.2).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 3.1 and 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. First article samples shall be inspected as specified in table II (see 3.1 and 6.2). First article inspection deficiencies shall be evidence that all samples produced prior to completion of first article inspection are similarly deficient, unless satisfactory evidence shows that all deficiencies have been corrected by the contractor.

TABLE II. Classification of inspections.

Title	Requirement	Inspection	First Article	Conformance	
				Examination	Acceptance tests
Materials and construction defects (see table III)	3.2 thru 3.2.3, 3.3 thru 3.3.2, 3.6, and 3.7	4.4.1 4.4.2	X X	X	
Insulation resistance	3.4.1	4.4.3	X		X
Dielectric strength	3.4.2	4.4.4	X		X
Output voltage	3.4.3	4.4.5	X		X
Phasing	3.4.4	4.4.6			X
Temperature	3.5.1	4.4.7.1	X		
Altitude	3.5.2	4.4.7.2	X		
Humidity	3.5.3	4.4.7.3	X		
Salt fog	3.5.4	4.4.7.4	X		
Sand and dust	3.5.5	4.4.7.5	X		
Fungus	3.5.6	4.4.7.6	X		
Chemical resistance	3.5.7	4.4.7.7	X		
Vibration	3.5.8	4.4.7.8	X		
Gun-firing shock	3.5.9.1	4.4.7.9.1	X		
Ballistic shock	3.5.9.2	4.4.7.9.2	X		
Basic shock	3.5.9.3	4.4.7.9.3	X		
Submergence	3.5.10	4.4.7.10	X		

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4.3 Conformance inspection.

4.3.1 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ANSI/ASQ Z1.4. Each sample selected shall be inspected as specified herein for the defects listed in table III. Conformance to a lot shall be accepted when zero (0) defects are found and rejected when one (1) or more defects are found.

4.3.2 Examination. Each sample selected according to 4.3.1 shall be inspected as specified in table II and shall be free from any defect as listed in table III.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major:		
101	Assembly, incomplete (see 3.3.1).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3 and 3.3.1).	SIE 1/
103	Weight not as specified (see 3.3.2).	Scale
104	Workmanship, faulty affecting performance (see 3.7).	Visual
Minor:		
201	Dimensions not affecting interchangeability not within tolerance (see 3.3.1).	SIE
202	Marking, improper (see 3.6).	Visual
203	Workmanship, faulty affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.3.3 Classification of defects. For examination purposes, defects shall be classified as listed in table III.

4.3.4 Acceptance tests. Each pickup shall be subjected to the acceptance tests specified in table II.

4.4 Methods of inspection.

4.4.1 Materials. Conformance to 3.2 thru 3.2.3 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.4.2 Defects. Conformance to 3.3 through 3.3.2, 3.6 and 3.7 shall be determined by examination for the defects listed in table III. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

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4.4.3 Insulation resistance. To determine conformance to 3.4.1, measure the insulation resistance between the coil and case using an applied potential of 50 V rms.

4.4.4 Dielectric strength. To determine conformance to 3.4.2, 100 V rms shall be applied between the coil and the case.

4.4.5 Output voltage. To determine conformance to 3.4.3, measure the output voltage under the conditions specified in table I using a standard test ring.

4.4.6 Phasing. To determine conformance to 3.4.4, the pickup shall be connected to a dc voltmeter with the positive lead connected to pin A and the negative lead connected to pin B. When the tip of the pickup is approached by a magnetic object, the metal needle shall deflect. The connections shall be reversed and the tip again approached with a magnetic object. There shall be no needle deflection.

4.4.7 Environmental tests. The pickup shall be subjected to the environmental tests specified herein. After completion of each environmental test, the performance tests specified in 4.4.3 and 4.4.5 shall be repeated.

4.4.7.1 Temperature. To determine conformance to 3.5.1, the pickup shall be tested as follows:

a. Accelerated aging. Dry the pickup in an air oven and subject it to a temperature of 204 °C (400 °F) for 168 hours.
b. High temperature. Upon completion of the accelerated aging, subject the pickup to a temperature of 100 °C (212 °F) for 100 hours.
c. Room temperature. Upon completion of the high temperature test, subject the pickup to a temperature of 25 °C (77 °F) for 8 hours.
d. Low temperature. Upon completion of the room temperature test, subject the pickup to a temperature of -54 °C (-65 °F) for 30 hours.

4.4.7.2 Altitude. To determine conformance to 3.5.2, the pickup shall be tested as follows. Subject the pickup to an atmospheric pressure of 31.19 inches of mercury (396 m (1300 feet) below sea level). Subject the pickup to an atmospheric pressure of 20.01 inches of mercury (3048 m (10,000 feet) above sea level).

4.4.7.3 Humidity. To determine conformance to 3.5.3, the pickup shall be tested as follows: subject the pickup to an atmosphere of clear water vapor for a period of 15 days under the conditions and cycle as specified in table IV. Immediately after 12 hours of the first test cycle, check the pickup for proper operation with the maximum specified voltage. During the last cycle of the 15-day test, operate the pickup throughout its entire range of operation under each of the ambient conditions specified. At the end of the 15-day test, dry the pickup without forced convection for 12 hours in air at not greater than 21 °C (70 °F) with not less than 50% relative humidity. At the conclusion of this period, check the insulation resistance and output

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voltage for conformance to specification requirements. After this, disassemble and inspect the pickup.

TABLE IV. Humidity conditions test cycle.

Time period (hours)	Relative humidity (%)	Ambient temperature °C (°F)
8	95±5	54±3 (129±5.4)
4	Holding chamber moisture content constant	54±3 (129±5.4) gradually down to
8	Holding chamber moisture content constant	21±3 (70±5.4)
4	Up to 95±5, holding chamber moisture content or adding vapor to reach required value	21±3 (70±5.4) uniformly up to 54±3 (129±5.4)

4.4.7.4 Salt fog. To determine conformance to 3.5.4, the pickup shall be tested as follows: Expose the pickup to a salt fog for not less than 48 hours. The fog solution shall be 5% sodium chloride (NaCl) and 95% distilled water by weight. The temperature shall be maintained between 32 °C (90 °F) and 35 °C (95 °F). Fog density shall be 3 quarts of solution per 28.31 m^3 (10 ft^3) of chamber volume per 24 hours.

4.4.7.5 Sand and dust. To determine conformance to 3.5.5, the pickup shall be tested as follows: Subject the pickup to a sand and dust test in accordance with MIL-STD-810, method 510. The pickup need not be operated during the test; however, at the conclusion of the test, and prior to any cleaning of the pickup, check insulation resistance and output voltage for conformance to specification requirements. Criteria for passing the test shall be as defined in MIL-STD-810, method 510.

4.4.7.6 Fungus. To determine conformance to 3.5.6, the pickup shall be tested as follows: Inoculate the pickup with a spore suspension and test in accordance with the requirements of MIL-STD-810, method 508. The temperature shall range from 27 °C (80 °F) to 29 °C (84 °F) and the relative humidity from 96 to 100% for a period of 28 days.

4.4.7.7 Chemical resistance. To determine conformance to 3.5.7, the pickup shall be exposed to the chemicals as specified for not less than 48 hours.

4.4.7.8 Vibration. To determine conformance to 3.5.8, the pickup shall be tested as follows: When normally mounted and while powered, subject the pickup to vibrations of 5 g magnitude in each direction along each of 3 mutually perpendicular axes at a logarithmic sweep rate. The total cycling vibration time shall be equivalent to 20 minutes per axis from 5 to 2000 to 5 Hz with each 20-minute test period representing 1609 km (1000 miles) of vehicle operation. Conduct a resonance sweep to identify up to four of the most severe resonances and then accomplish resonance dwells for 1/6 of the total cycling vibration time at each resonance at the specified levels.

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4.4.7.9 Shock.

4.4.7.9.1 Gun firing shock. To determine conformance to 3.5.9.1, the pickup shall be tested as follows: When normally mounted and while powered, subject the pickup to 3 shocks of 100±10 g for 1.5±0.2 ms along each of 3 mutually perpendicular axes (a total of 18 shocks).

4.4.7.9.2 Ballistic shock. To determine conformance to 3.5.9.2, the pickup shall be tested as follows: When normally mounted and while powered, subject the pickup to 3 half sine wave shock pulses of 200±20 g along each of 3 mutually perpendicular axes (a total of 18 shocks).

4.4.7.9.3 Basic shock. To determine conformance to 3.5.9.3, the pickup shall be tested as follows: When normally mounted and while powered, subject the pickup to 3 half sine wave shock pulses of 40 g for 18 ms along each of 3 mutually perpendicular axes (a total of 18 shocks).

4.4.7.10 Submergence. To determine conformance to 3.5.10, the pickup shall be completely submerged in the fluids specified in 3.5.7. The duration of submergence shall be 2 hours to insure total submergence of the pickup. The pressure of the fluid shall be maintained at 345 kPa (50 psig).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pickup covered by this specification is intended to be used to generate signals reflecting the two rotor speeds within the AGT 1500 gas turbine engine of the M1 tank.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type (see 1.2)
c. If required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2 and 2.3).
d. When first article is required (see 3.1).
e. Identification and marking if other than as specified (see 3.6).
f. Inspection conditions if other than as specified (see 4.2).
g. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

AGT 1500 gas turbine engine
Rotor
Signals

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AT

Preparing Activity:
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